

**National Security Implications of Advanced Air Mobility (AAM): Cyber and
Physical Threats in an Evolving Airspace**

Devin Q. Tran

Department of Geography, Brigham Young University

Dr. David Simpson

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Abstract

The emergence of Advanced Air Mobility (AAM) represents an ongoing shift in the aviation landscape from the traditional hub-and-spoke model towards distributed networks consisting of vertiports, autonomous drones, and electric vertical takeoff and landing (eVTOL) aircraft. Advancements in AAM promise enhanced logistics capabilities, improved emergency response, and reduced ground-traffic congestion, while its decentralized nature presents unprecedented national security vulnerabilities. This paper will examine how a distributed network created by AAM technology expands the national threat landscape by converging aviation, critical infrastructure, and cybersecurity. Threats created by this unique cyber-physical relationship will need to be addressed through a comprehensive whole-of-government approach combining federal authority, state implementation, and local coordination to secure this new aviation ecosystem.

National Security Implications of Advanced Air Mobility (AAM): Cyber and Physical Threats in an Evolving Airspace

Long-standing aviation infrastructure characterized by large airports, security checkpoints, and controlled access represent the centralized nature of the hub-and-spoke model that has operated efficiently in the United States since the 1950s. Introduction of revolutionary Advanced Air Mobility (AAM) aircraft with capability to operate from diverse locations including vertiports, residential rooftops, and retail centers challenges this paradigm. Technological advancements of electric vertical takeoff and landing (eVTOL) aircraft, unmanned aircraft systems (UAS), and other advanced platforms seek to revolutionize urban transportation, logistics, and emergency response capabilities.

The shift towards distributed aviation networks introduces new variables and complexity creating unprecedented challenges for national security. Physical and procedural boundaries currently imposed by the Transportation Security Administration (TSA) and Federal Aviation Administration (FAA) fail to address the vulnerabilities created by restructuring traditional aviation infrastructure. As AAM infrastructure continues to develop interweaving with existing urban environments and critical systems, the nation will face an expanded threat surface that will require careful coordination and strategic security planning between federal, state, and local jurisdictions.

The Emerging AAM Landscape

The Federal Aviation Administration defines Advanced Air Mobility as "a rapidly-emerging, new sector of the aerospace industry which aims to safely and efficiently integrate highly automated aircraft into the National Airspace System (NAS). AAM is not a single technology, but rather a collection of new and emerging technologies being applied to the aviation transportation system, particularly in new aircraft types" (Federal Aviation Administration, 2024).

AAM technology aims to revolutionize traditional logistics processes by utilizing unmanned aircraft systems capable of completing direct-to-consumer deliveries and addressing the issue of last-mile distribution. Manna, an Irish drone delivery startup, has launched operations in Dallas, Texas completing successful drone deliveries (Dallas Innovates, 2023). Other major retailers including Amazon, Walmart, and UPS have initiated similar drone delivery programs, demonstrating the commercial application of these systems (Wolf, 2024). These operations showcase the potential future for logistics operations and distribution in urban and suburban environments.

Electric vertical takeoff and landing aircraft combine the hovering capabilities of helicopters with efficient forward flight, enabling their deployment as "air taxis" for passenger mobility while bypassing ground-based traffic congestion. On August 15, 2025, Joby Aviation completed the first piloted air taxi flight between two public airports traveling from Marina to Monterey, California (Joby Aviation, 2025). This successful operation demonstrated the

ability for eVTOL aircraft to operate safely within the existing national airspace system.

Infrastructure requirements of AAM differ fundamentally from traditional aviation. Vertiports and vertistops are facilities designed to enable eVTOL operations, integrating into existing urban infrastructure including, rooftops, parking structures, retail centers, and residential areas. Enhanced capabilities of eVTOL aircraft eliminate the need for long runway systems, allowing for flexible site selection for vertistop and vertiports to be constructed. Despite improving versatility, implementing a distributed network without the precautions standard to centralized aviation facilities creates new security vulnerabilities.

Platform advancements in AAM also offer improved capabilities for emergency and disaster response through rapid delivery of critical medical supplies, providing real-time data, and assisting with search-and-rescue efforts. NASA outlined the role that AAM will play in emergency response and disaster relief highlighting greater accessibility of difficult to reach locations, creating a “highway in the sky” that promotes safety and efficiency for first responders during critical events.

Physical Security Vulnerabilities

The introduction of AAM aircrafts into the aviation ecosystem presents new challenges for national security and public safety. Ease of access to highly capable drones opens the door for weaponization of drones. With “Prosumer

drones” costing as little as \$1,500, affordable drones can be used as terror devices carrying devastating payloads (Drone Pilot Ground School, 2024). The conflict between Ukraine and Russia has involved multiple large-scale drone attacks. Most notably in June 2025 “Operation Spiderweb” conducted by Ukrainian forces utilized 117 drones in an attack on multiple Russian air bases. The results of the strike were devastating with 41 Russian strategic aircrafts attacked and an estimated \$7 billion in damage on Russia (Institute for the Study of War, 2025).

While military attacks represent an extreme example of drone weaponization, similar capabilities could be used to target critical domestic infrastructure. Reliance on power generation facilities, water treatment plants, and other essential services represent high-value targets where even less coordinated strikes have the potential to cause damage on a national scale. AAM infrastructure itself creates challenges as hydrogen storage facilities, rooftop charging stations, and other necessary facilities create additional vulnerabilities within greater infrastructure networks.

The National Safe Skies Alliance identified AAM operation vulnerabilities including a lack of employee and passenger vetting, electrical infrastructure and battery safety concerns, insufficient screening of persons and baggage, protocols for managing unruly passengers, and cybersecurity deficiencies (National Safe Skies Alliance, 2021). Each vulnerability highlights a potential vector for security breaches or unsafe conditions.

Traditional airports maintain safety and security through utilizing sterile area boundaries protected by security screening checkpoints. Physical security concerns arise from eVTOL aircraft arrival from unregulated vertiports without strict security screening for passengers and cargo. The security screening problem does not end there, concerns regarding emergency landing of eVTOL vehicles present similar security threats. The absence of prior security screening of conventional airports creates uncertainties for any AAM aircraft that departed from an unregulated facility seeking to land at a conventional airport, compromising established security perimeters and sterile areas.

Rapid expansion of AAM operations has outpaced development of corresponding security regulations. TSA is in the process of developing “robust frameworks to ensure security without stifling progress,” but has not yet released comprehensive regulations addressing AAM-specific security requirements (Transportation Security Administration, 2024). The result is a regulatory gap with immediate concerns for national security across multiple dimensions.

Cybersecurity Threats

AAM aircraft function as “flying computers” that require constant data links (C2, 2025). These communication-based systems are vulnerable to hacks, jamming, spoofing, ransomware, or any other cyber-attacks. A successful cyber attack could potentially ground entire fleets, cause aircrafts to deviate from

planned routes, or result in crashes. With the right technology the wrong actor could conduct targeted attacks using AAM aircraft for devastating effects on critical infrastructure.

The 2021 Colonial Pipeline ransomware attack highlighted the vulnerability of critical infrastructure to cyber threats, resulting in a nationwide fuel supply crisis, mass panic, and economic consequences (Cybersecurity and Infrastructure Security Agency, 2023). Fearing another incident, the federal government implemented the National Cybersecurity Strategy (2023), tasking the TSA with the development of mode-specific cybersecurity standards.

Current cybersecurity standards remain insufficient for managing complexities of AAM systems and the Government Accountability Office is pushing for stronger standards for both aviation and automated aircraft platforms. Understanding the critical role that comprehensive cybersecurity standards hold in national security, the SAE G-32 Cyber Physical Systems Security Committee is working to standardize cybersecurity practices across sectors (SAE International, 2024).

Aircraft system design necessitates proper cybersecurity as improper “bundling of systems” or integrating critical flight controls, engine management, and secure communication with non-critical components creates additional vulnerabilities. Components sourced from a compromised supplier that are mixed with essential systems create backdoors or vulnerabilities that can be used to gain unauthorized access to critical controls. A company using

components from an insecure supplier in their drone fleet could have their entire fleet grounded or hijacked.

Cybersecurity challenges are compounded by economic constraints of AAM startups. Aviation pioneers that are struggling to see commercial success often lack the resources required for developing robust cybersecurity measures. Without cybersecurity requirements, motivation for cyber resilience suffers creating vulnerabilities for weak systems. This vulnerability represents a broader national security threat imposed by the emerging AAM landscape.

The Governance Gap

National security has traditionally operated as a federal responsibility with government agencies in charge of varying aspects of security. The FAA regulates airspace and aircraft operations while the TSA helps secure airports by screening passengers and cargo, and the Department of Defense is responsible for addressing major aerial threats to national security. This division of responsibilities effectively distributes authority and ensures security within traditional aviation's centralized infrastructure.

The distributed network created by AAM's location versatility challenges this security framework. Construction, operation, and ownership of vertiports relies on multiple stakeholders including, commercial operators, private developers, and municipal governments. Without the consolidated security measures and clear jurisdictional authority of conventional airports, the

distributed network lacks a robust framework. Vertiports can be embedded into existing urban infrastructure, raising questions surrounding safety, oversight, privacy protections, and noise regulation for communities.

This governance gap presents significant security vulnerabilities. Federal authorities maintain jurisdiction over airspace operations but lack authority to address ground-based infrastructure and local security. State and local governments frequently lacking resources, expertise, and statutory authority will be responsible for addressing the complexities of ground-based infrastructure and local security protocols. Jurisdictional fragmentation results in unclear responsibility distribution and creates both cyber and physical vulnerabilities for vertiport locations. Attacks on multiple vulnerable locations could overwhelm local response capabilities and exceed federal jurisdiction. Absence of clear command structure and response protocols increases risk of critical security breaches.

Mitigation Strategies

“Zero trust” is a familiar model within the cybersecurity community, this model offers a robust framework for securing AAM system architecture. Operating on the principle that no entity inside or outside of the network should be automatically trusted, this model protects critical data even in the event of a network compromise (Cybersecurity and Infrastructure Security Agency, 2024). Applications of zero trust architecture in AAM would elicit encrypted

communications for all data links, continuous user and system authentication, real-time monitoring, and automated isolation for compromised systems.

Creating a layered defense through implementing zero trust principles creates a stronger system where attackers will be forced to overcome multiple security controls, increasing the difficulty and cost of successful attacks while creating more opportunities for early detection and response.

Taking a proactive approach to threat detection is essential to AAM security. Companies such as Hidden Level have developed powerful technology for monitoring, detecting, and identifying unauthorized drones using radio frequency (RF) sensing. Hidden Level focuses on protecting public venues and infrastructure, supporting emergency operations, and safe airspace monitoring through real-time data (Hidden Level, 2024).

Incorporating advanced threat detection systems at vertiports, critical infrastructure, and high-value targets provides early warning of potential security breaches. Combining enhanced detection capabilities with robust response protocols enables rapid intervention before threats can materialize into actual attacks. Partnerships between defense technology companies and local AAM operators represent a practical approach towards implementing distributed threat detection and creating a safer AAM ecosystem.

Designing resilient physical infrastructure can help reduce vulnerability to attacks. Vertiports created with fire suppression systems designed for battery fires, isolated battery storage, redundant power supplies, and strategic

placement of critical systems will minimize vulnerability to tampering or targeted attacks. Building defense into AAM infrastructure helps to lower risk of single-point failures or isolated attacks impacting the broader system. Strengthening individual facilities through resilient infrastructure helps promote security across the distributed AAM network.

Policy and Regulation

Addressing the policy and regulatory gaps in AAM security will require clear federal leadership. AAM security will need to be designated as an explicit federal authority by Congress so that appropriate authority can be granted to the TSA or another agency with explicit authority. The authority should include funding and resources to develop and enforce security standards for vertiports, fleet operators, and supply chains.

Federal agencies will need to develop comprehensive security standards applicable for all AAM operations. Standards addressing physical security requirements, cybersecurity protocols, personnel vetting procedure, cargo screening, and emergency response protocols. Standardizing security ensures consistent practices across the distributed AAM ecosystem while providing clear compliance requirements for operators and local jurisdictions.

Compliance to cybersecurity standards needs to be demonstrated by AAM operators and platform manufactures prior to receipt of FAA operation approval. Certification should include independent security audits, penetration

testing, supply chain security analysis, and ongoing security monitoring. Integrating cybersecurity certification into the operation approval process ensures that security is proactive and multi-faceted.

Coordination on a state and local level is imperative for ensuring that federal security standards are implemented in AAM deployment. The federal government should encourage and incentivize regional security task forces modeled on existing frameworks such as those developed by the North Central Texas Council of Governments (NCTCOG). Task forces can work as coordination hubs aligning local AAM implementation with national security requirements and applying federal guidelines on a regional scale. Information sharing between federal agencies, state governments, local authorities, and private operators will be improved by task force coordination creating greater threat awareness and response capabilities.

Conclusion

Advanced Air Mobility represents a fundamental restructuring of how aircrafts integrate with society. AAM promises enhanced logistical capabilities, emergency response, and freedom from ground-based traffic. Realization of this promise depends on carefully addressing complex national security challenges created by AAM's distributed network and local infrastructure requirements.

While traditional aviation security relies on checkpoints, controlled spaces, and oversight from federal agencies, AAM operations distributed across various nodes and reliant on local infrastructure cannot be secured by adopting these traditional methods. Convergence of physical threats including weaponization of drone technology, breach of sterile areas, screening concerns and cyber threats a threat landscape of unprecedented complexity.

Securing the AAM ecosystem demands a whole-of-government approach relying on powerful federal leadership, state implementation, and local enforcement. Federal agencies will need to create comprehensive security standards and coordinate national threat responses. State and local governments will be responsible for protecting the distributed infrastructure, ensuring compliance with federal security requirements. Efforts for coordination should be supported by federal funding of security task forces established for improving regional coordination and alignment with national security objectives.

Industry stakeholders will play a critical role in designing AAM systems with secure architecture and developing threat mitigation technologies. However, economic pressures facing AAM startups may constrain security investments necessitating federal funding or requirements for security measures.

AAM infrastructure will need to be understood as not just flying vehicles but nodes within a complex distributed network with direct ties to national

security. The security of this infrastructure relies on recognizing complex relationships between aviation operations, critical infrastructure, cybersecurity, and urban landscapes. A comprehensive security architecture will be required for mitigating vulnerabilities in this emerging ecosystem.

The challenge of building a new aviation security paradigm to fit AAM's distributed architecture is substantial. However, promises of enhanced logistical capabilities, emergency response, and urban transportation represent transformative societal benefits that justify investment in developing comprehensive security frameworks and government coordination. Advanced Air Mobility is set to transform how society moves, works, and connects. The real question now is whether we will shape that future with proactive, integrated security measures or reactively in response to actual security incidents.

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